Capstone Evaluate and Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Recall and choose your starting point

> Record your starting explanation for Capstone Evaluate and Improve before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone Build

Without opening the old lesson, explain one key idea from Capstone Build. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Human AI Workflow

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Copyright and Provenance

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

Complete the supported Capstone Evaluate and Improve trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone evaluation and revision. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Capstone Evaluate and Improve explanation against the supplied facts before discussing it.

Use the worked case for Capstone evaluation and revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Capstone Evaluate and Improve target to a trace and a justified conclusion.

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Capstone Evaluate and Improve case and show the deciding evidence.

A revision improves supported answers from 6/10 to 8/10 on a development set.

What belongs in the report?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A revision improves four factual questions but fails two previously successful abstention cases. What should the evaluation report?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Capstone Evaluate and Improve adds to Verified Study Buddy and keep the evidence.

Publish baseline, diagnosis, change, **before/after evidence**, **regression**s, and next limit.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Capstone Evaluate and Improve facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Capstone Evaluate and Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Trace the worked case

> Trace the worked case. Apply the supplied Capstone Evaluate and Improve facts and keep your reasoning.

On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Audit the limit

> Audit the limit. Apply the supplied Capstone Evaluate and Improve facts and keep your reasoning.

A report omits this qualification: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Capstone Evaluate and Improve facts and keep your reasoning.

Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

My reasoning and evidence

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Capstone Evaluate and Improve facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Capstone Evaluate and Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

Latency cost and resource use — W1 Explain the mechanism

> Use the supplied Capstone Evaluate and Improve task to record your reasoning, evidence and next step.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 18 OF 26

Latency cost and resource use — W2 Trace the worked case

> Use the supplied Capstone Evaluate and Improve task to record your reasoning, evidence and next step.

A request uses 500 **input** units and 200 **output** units; starts at 10.0 seconds and finishes at 13.5. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 19 OF 26

Latency cost and resource use — W3 Audit the limit

> Use the supplied Capstone Evaluate and Improve task to record your reasoning, evidence and next step.

A report omits this qualification: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 20 OF 26

Latency cost and resource use — W4 Apply a new case independently

> Use the supplied Capstone Evaluate and Improve task to record your reasoning, evidence and next step.

Request X uses 1000 **input** and 300 **output** units; Y uses 200 **input** and 100 **output**. Under the same fictional **tariff** compute individual and total cost.

My reasoning and evidence

WORKBOOK · 21 OF 26

Latency cost and resource use — W5 Design a revealing follow-up

> Use the supplied Capstone Evaluate and Improve task to record your reasoning, evidence and next step.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Capstone Evaluate and Improve

> Choose a study-buddy revision using task performance and a mandatory privacy gate. Explain the labelled step before checking the changed condition.

Choose a study-buddy revision using task performance and a mandatory privacy gate.

On eight fixed validation cases, baseline passes 5 and candidate passes 7. One candidate response includes a restricted teacher note; privacy requires zero such exposures.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/35

Capstone Evaluate and Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare totals

Baseline: 5/8. Candidate: 7/8.



2. Apply privacy gate

The restricted-note exposure fails the mandatory release condition.



3. Revise and retest

Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

CHANGE ONE THING

A repaired candidate passes 6/8 with zero restricted-note exposures.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose a study-buddy revision using task performance and a mandatory privacy gate.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

1. Revise and retest

2. Compare totals

3. Apply privacy gate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Revise and retest Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Read the picture in order.

Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A repaired candidate passes 6/8 with zero restricted-note exposures. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. It passes this privacy gate and its validation task rate is 75%.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It passes this privacy gate and its validation task rate is 75%. A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Repair a missing building block

Review Capstone Build (ai-3/34). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: What additional numbers would a half-water claim require?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Copyright and Provenance (ai-3/29). Goal: Explain the **mechanism** and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

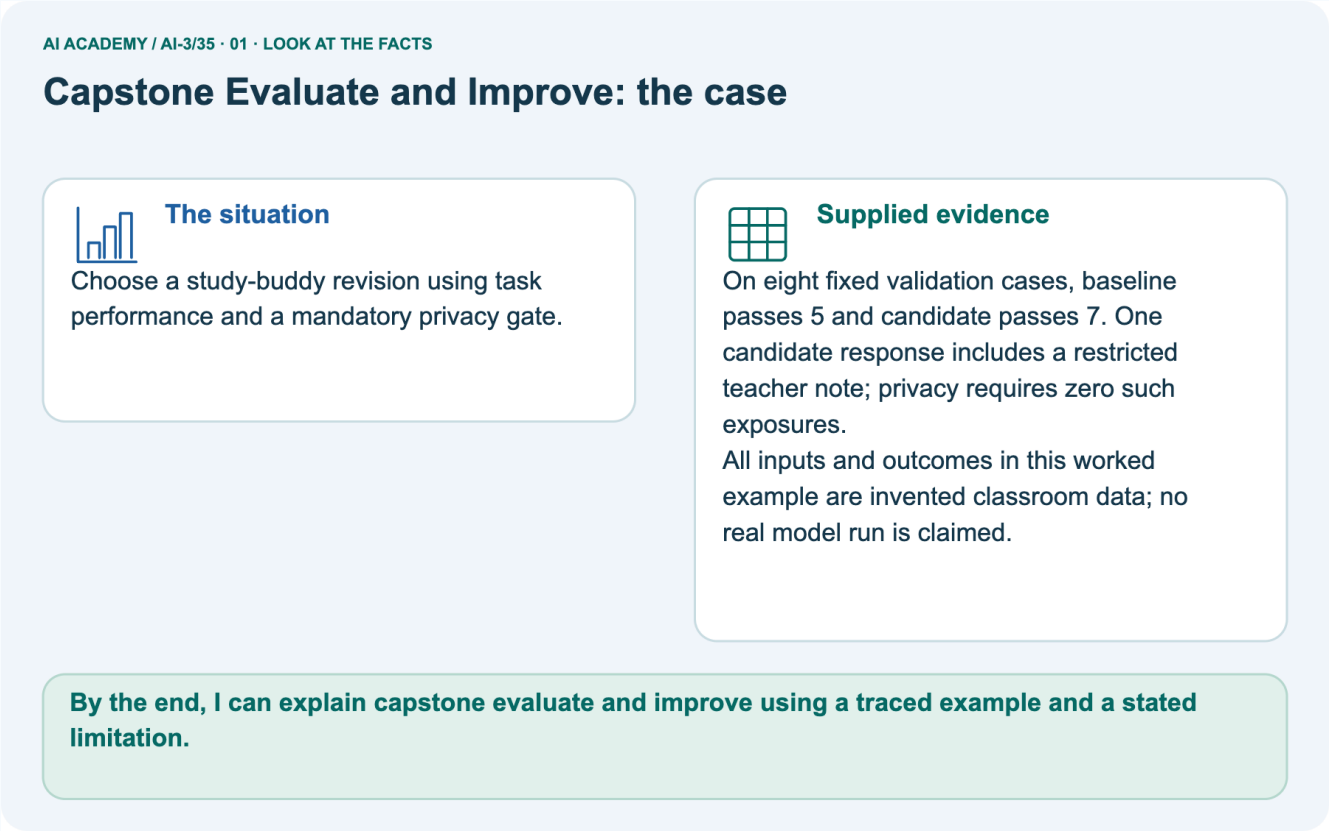
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Capstone Evaluate and Improve picture from its labels, then record a relevant limit.

01. Capstone Evaluate and Improve: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

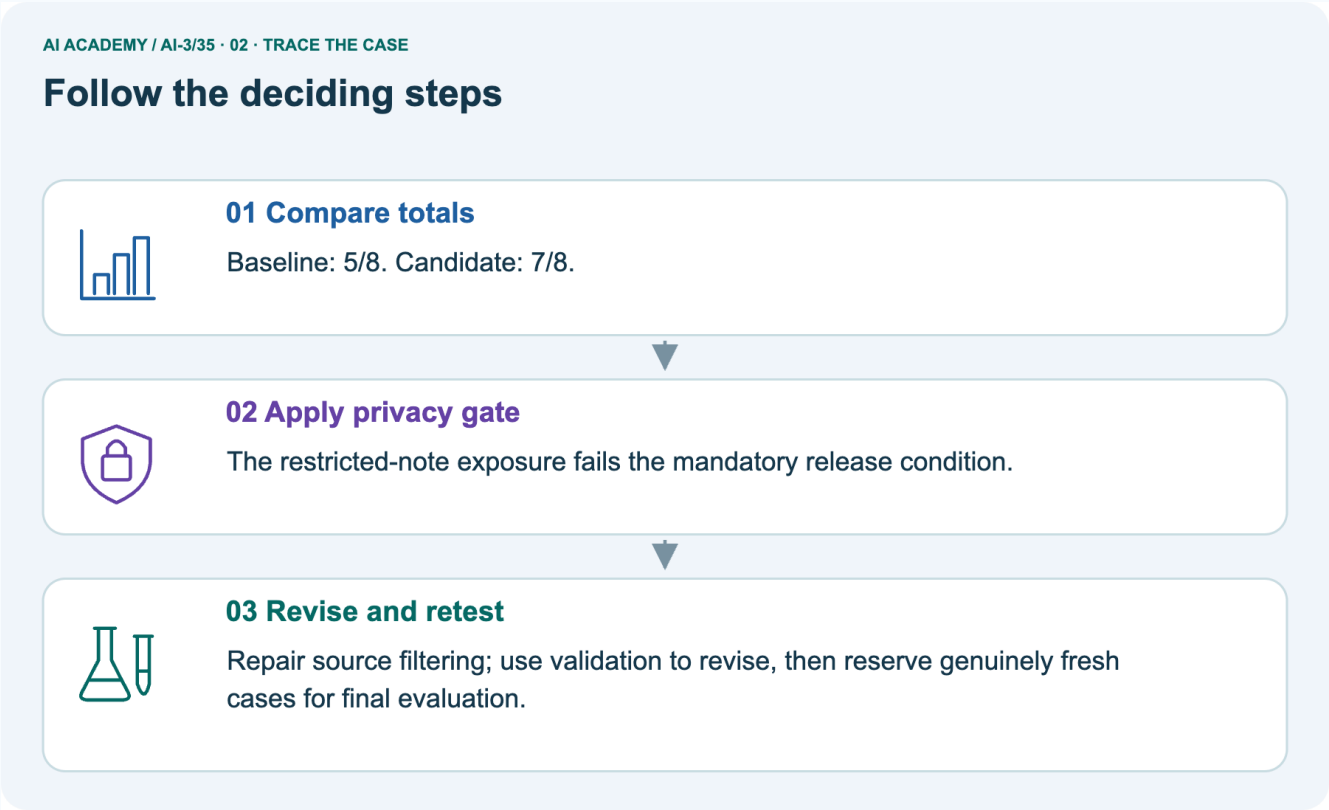
Choose a study-buddy revision using task performance and a mandatory privacy gate. Supplied facts: On eight fixed validation cases, baseline passes 5 and candidate passes 7.

One candidate response includes a restricted teacher note; privacy requires zero such exposures. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

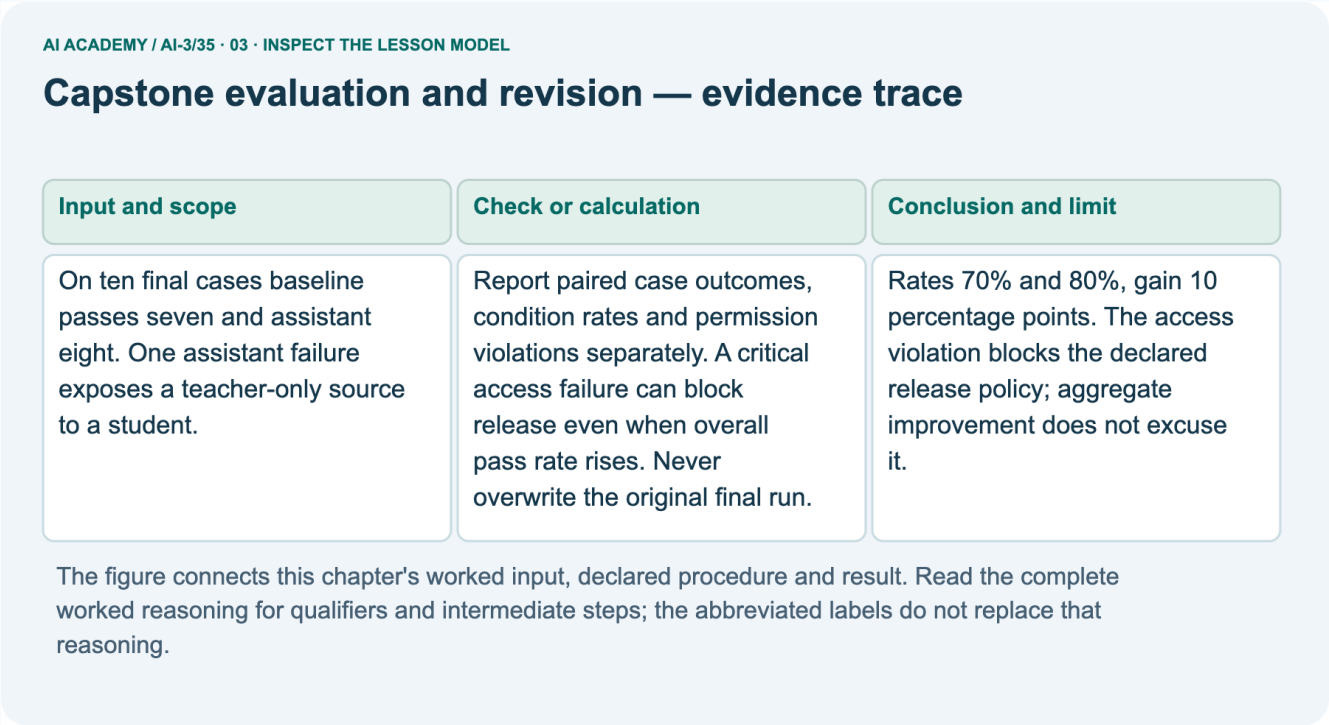
Compare totals: Baseline: 5/8.

Candidate: 7/8. → Apply privacy gate: The restricted-note exposure fails the mandatory release condition. → Revise and retest: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone evaluation and revision — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone evaluation and revision — **evidence** trace. On ten final cases baseline passes seven and assistant eight.

One assistant failure exposes a teacher-only source to a student. | Report paired case outcomes, condition rates and permission violations separately.

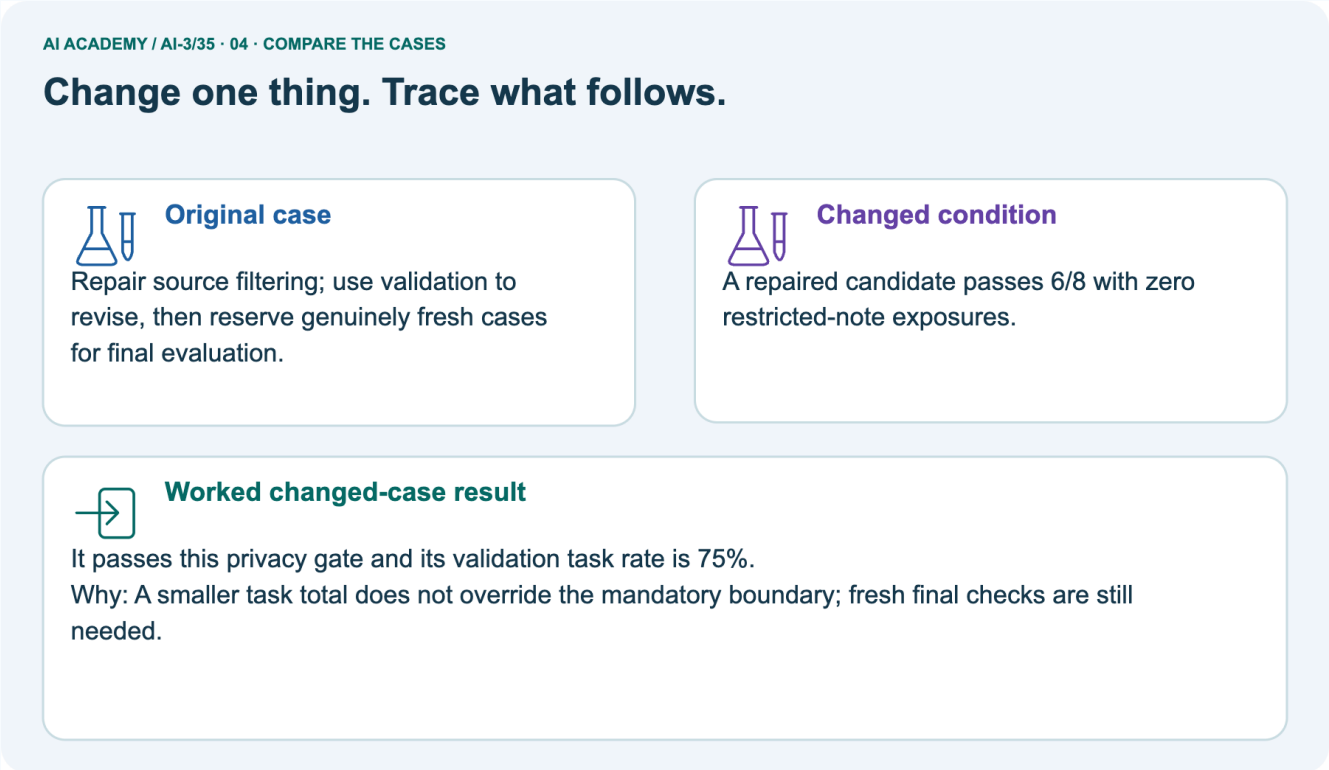
A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run. | Rates **70%** and **80%**, gain 10 percentage points.

The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

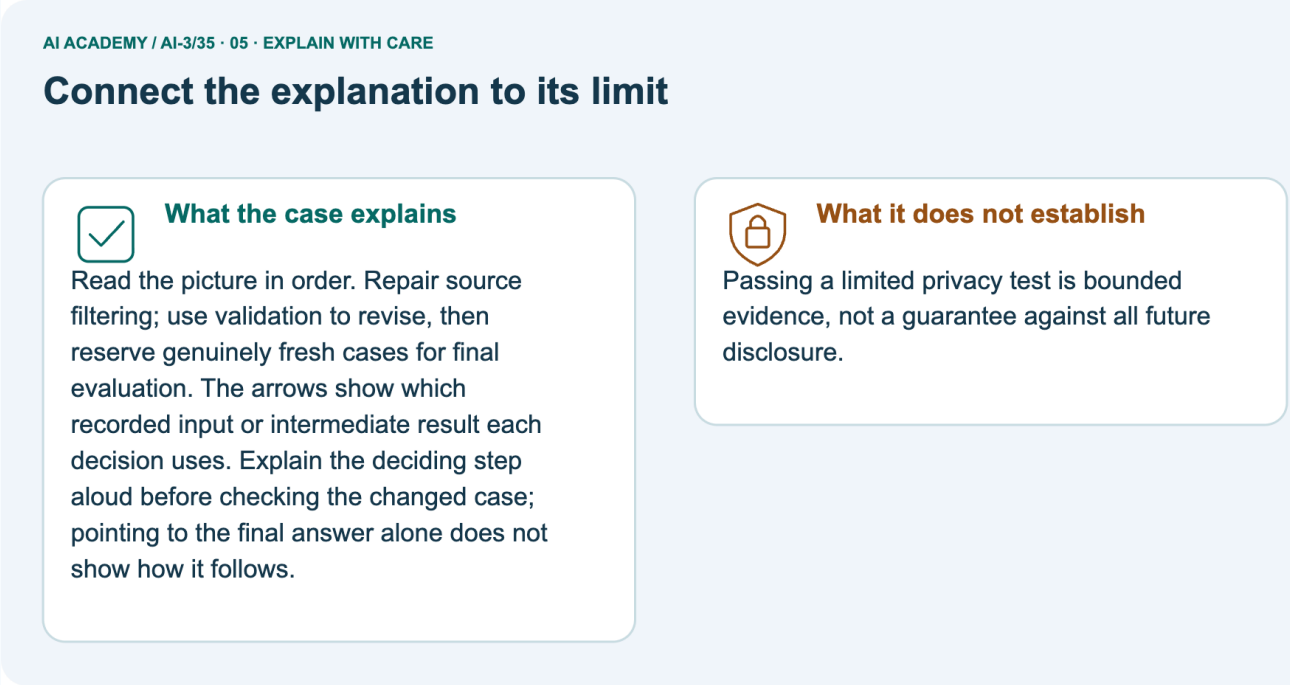
Original case: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Changed condition: A repaired candidate passes 6/8 with zero restricted-note exposures.

Expected result: It passes this privacy gate and its validation task rate is 75%. Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

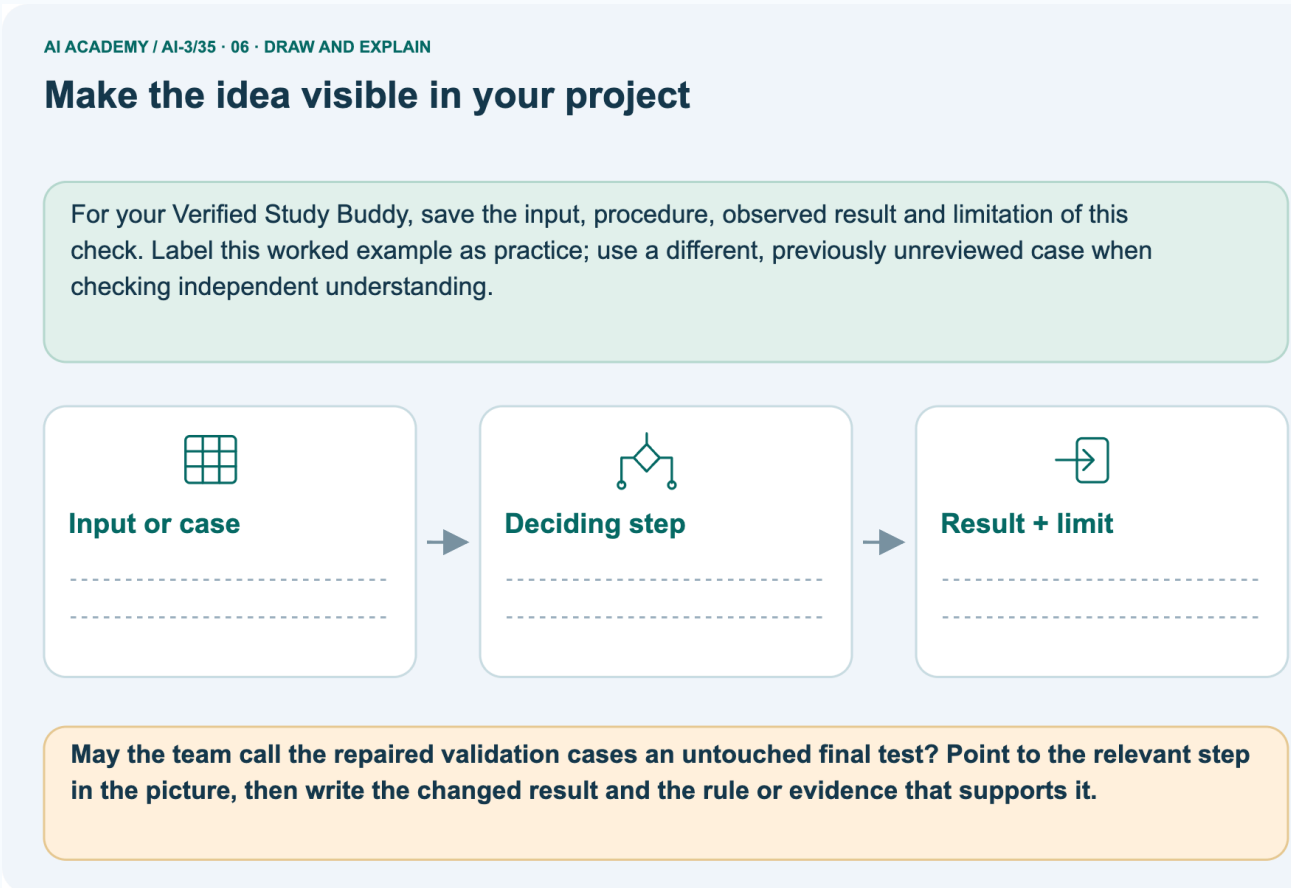
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.